

A/B Testing

Claim: A company that make protein bars claims that 15gms of protein in every bar.



Sample (500)
↓
SM = 14.98 gms

PM vs SM
15gms

Step 1: Writing Null & Alternative Hypothesis

Null Hypothesis (H_0): Default assumption or an assumption where we say that nothing has changed.

NH: The avg protein in a protein bar = 15gms

NH contains: =, $\geq$, $\leq$
↓ ↑ ↑
(PM = SM) at least min at most max

Alternative Hypothesis (H_1): Alternate assumption or an assumption that we have in our mind, we want to test.

H_0 : The avg protein in a protein bar $\neq 15\text{gms}$

H_0 contains: $\neq, >, <$

$S_{PM} < P_{PM}$
 $S_{PM} \neq P_{PM}$

② Objective: Try to reject the H_0 (always)

③ Decide the significance level (α):

How much risk of being wrong are you comfortable with.

$$\begin{aligned}\text{Significance level} &= 1 - \text{confidence level} \\ &= 1 - 0.95 \\ &= 0.05 (5\%)\end{aligned}$$

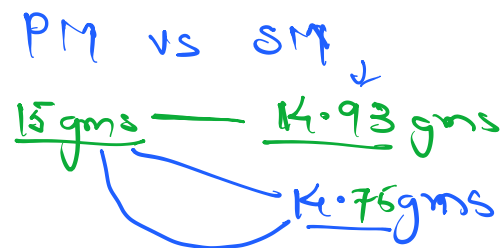
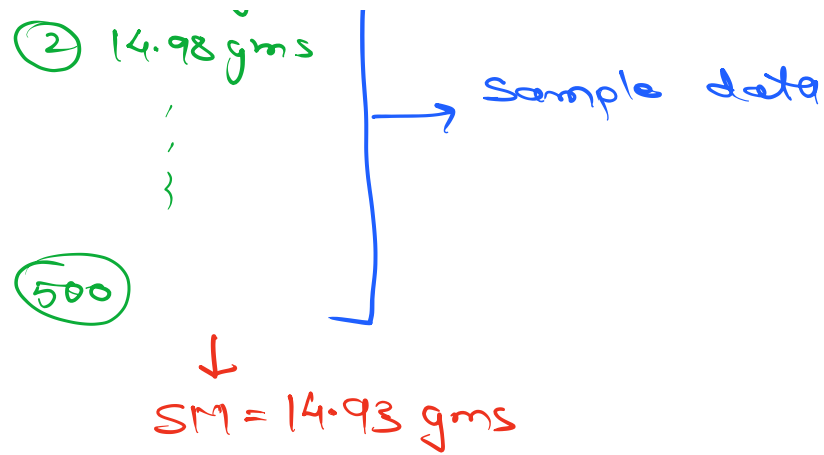
Most used values are: 1%, 5% & 10%.

④ p-value $\rightarrow$ considered as our final evidence

PM vs SM
15gms ?

Ex: sample of 500 protein bars

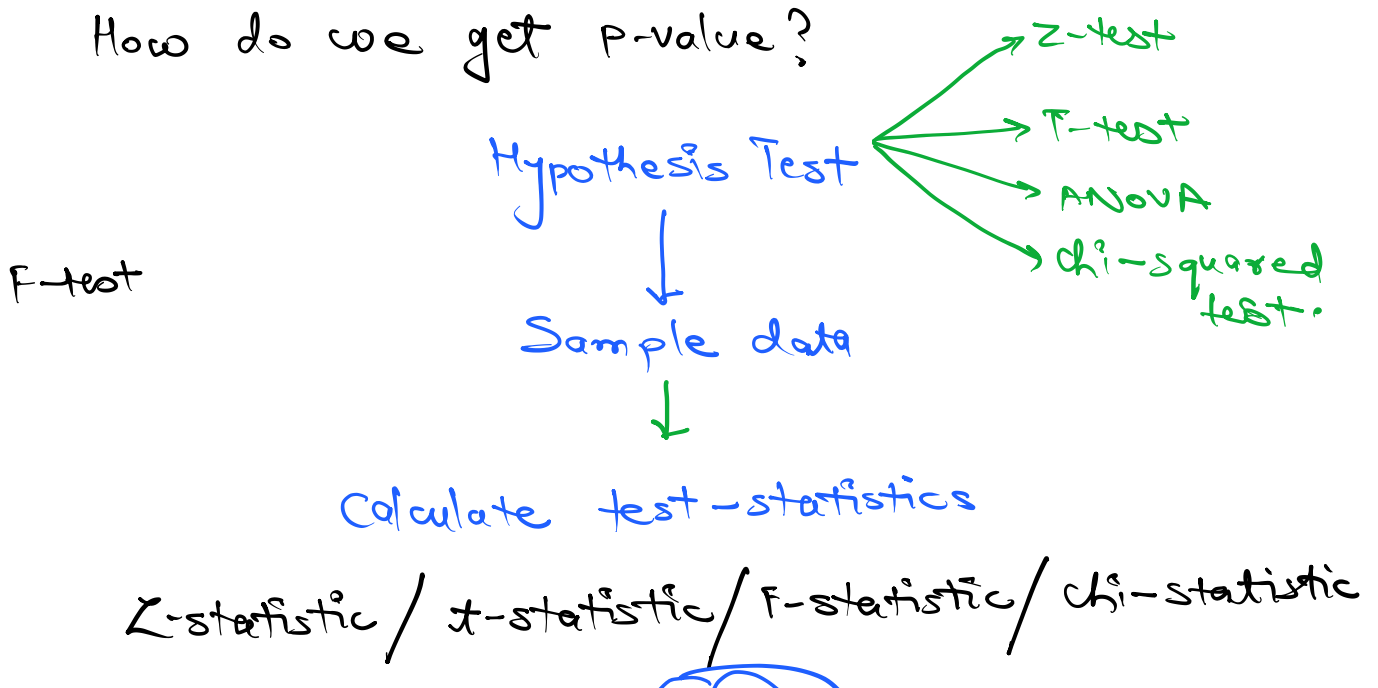
sample size $\rightarrow$ 500 protein bars
 $\rightarrow$ ① 15.01gms



p-value will help us decide whether we should H_0 or not.

It indicates the probability of obtaining your observed results (or more extreme results) if the null hypothesis is true.

How do we get p-value?



$$t\text{-stat} = \frac{(SRQ - PM)}{SD / \sqrt{SS}} = \text{number}(-, t, 0)$$

test-statistic value tells how different the PM and calculated SRQ are from each other, in terms of SDs.

higher the test-statistic $\rightarrow$ higher the chances of rejecting H_0 .



Once we have the test-statistic value, we use it to calculate (p-value).

(p-value)

Higher the test-statistic $\rightarrow$ smaller the p-value



Higher the chances of rejecting H_0 .

p-value (0-1)
0.03 (3%)

p-value says:

Acc to the evidence, what is probability of H_0 being True.

[97% $\rightarrow$ prob of H_0 being false (against H_0)
3% $\rightarrow$ prob of H_0 being True (favour of H_0)]

⑤ If

$p\text{-value} < \text{significance level } (\alpha)$



Reject the H_0 .

⑥ Summarize

Hypothesis Test



Sample Data



cal test-statistic



cal. p-value



if $p\text{-value} < \frac{1\%}{SL(\alpha)}$

